

Can a Small LSTM Beat *HAR*?

Honest measurement of a deep-learning edge in daily, range-based volatility forecasting — and where that edge does, and doesn't, come from.

HAR's reputation was earned on data we don't have

- HAR (Corsi, 2009): three OLS terms on daily / weekly / monthly lagged volatility — famously **hard to beat**.
- That reputation was earned on **high-frequency realized variance**.
- This project has only daily OHLC — so the targets are daily **range-based** variance proxies (Rogers–Satchell + overnight; Garman–Klass), a distinct estimator family.
- The real question: does HAR's dominance transfer to this regime — and can anything beat it here?

A protocol built to make self-deception hard

- **Frozen snapshot** — SPY + VIX, SHA-256 checksummed, pinned end date 2026-06-30.
- **Chronological walk-forward** — train 2005–17, val 2018–19, test 2020–2026, monthly refit (78 folds).
- **Hyperparameters tuned once** (50-trial Optuna) and frozen before any test-region contact.
- **10-seed ensembles**, Diebold–Mariano (HLN-corrected) inference — edge requires $p < 0.05$ and $\geq 80\%$ seed agreement.

HAR earns its reputation here

MODEL	TV H=1	TV H=5	OC H=1	OC H=5
Persistence	1.0239	0.4275	0.6803	0.3904
EWMA	0.5597	0.3654	0.4756	0.3483
AR(1) log	0.5728	0.3476	0.4290	0.3048
GARCH(1,1)	0.4822	0.2993	0.4731	0.3410
HAR (log)	0.4766	0.2877	0.3805	0.2677

QLIKE, lower is better. HAR beats every classical model significantly — **except GARCH on TV**, where the two are statistically indistinguishable ($p = 0.64 / 0.37$).

Architecture alone finds nothing

TV TARGET	LSTM (RV-ONLY)	HAR	DM P	SEED AGR.
h=1	0.4726	0.4766	0.585	60%
h=5	0.2917	0.2877	0.667	90%

A genuine null — not a failure. Given only the information HAR already uses, the network recovers **HAR-level performance and nothing more**. The protocol was equally prepared to publish this.

Auxiliary features reverse the null — decisively

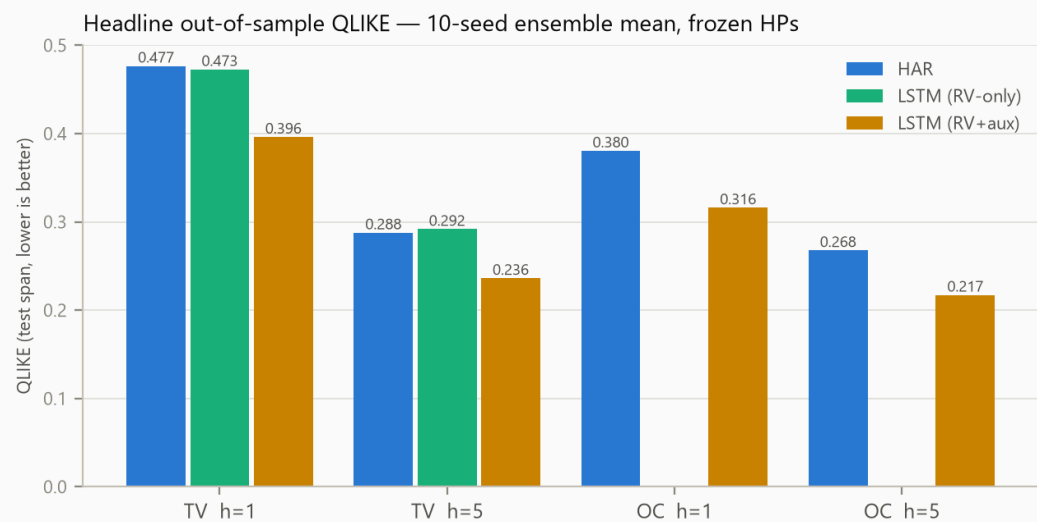


FIG. 1 — HEADLINE OUT-OF-SAMPLE QLIKE, RV+AUX LSTM VS. HAR

17–19%

QLIKE improvement over HAR, every block

4.9×10^{-8} – 9.7×10^{-5}

DM p-values, all four target × horizon blocks

100%

seed agreement, every block

Features, not architecture

- RV+aux beats RV-only **head-to-head** (TV $h=1$ $p = 7.2 \times 10^{-9}$)
— the edge traces to the auxiliary features, not the sequence model.
- Present at **both horizons** — nearly the same relative size at $h=5$ despite reused, unre-tuned hyperparameters.
- Present on **both targets** — unchanged whether overnight variance is included (TV) or excluded (OC).
- Present in **every regime** — COVID, calm 2021, the 2022 bear, and the 2023–26 recovery. The cumulative edge never gives any of it back.

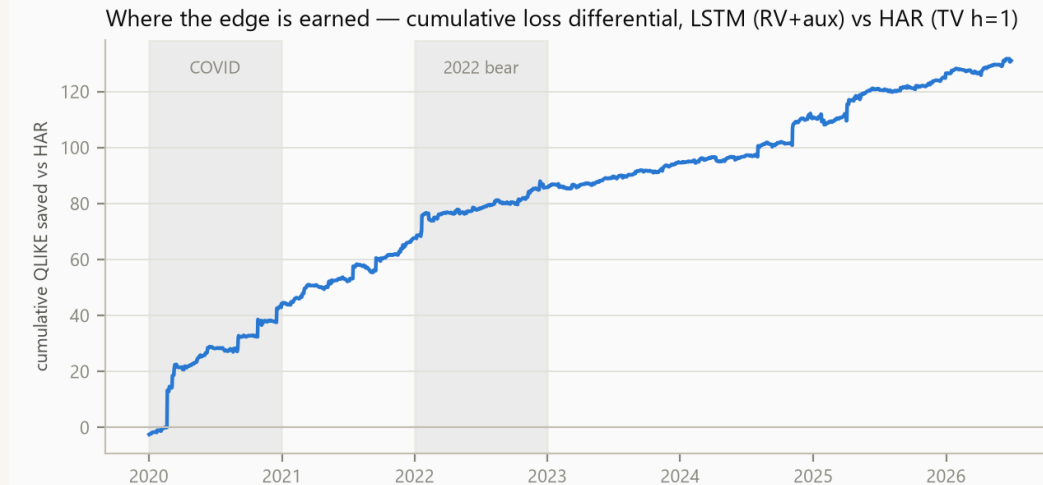


FIG. 2 — CUMULATIVE QLIKE DIFFERENTIAL, TEST SPAN 2020–2026

The network is data-hungry

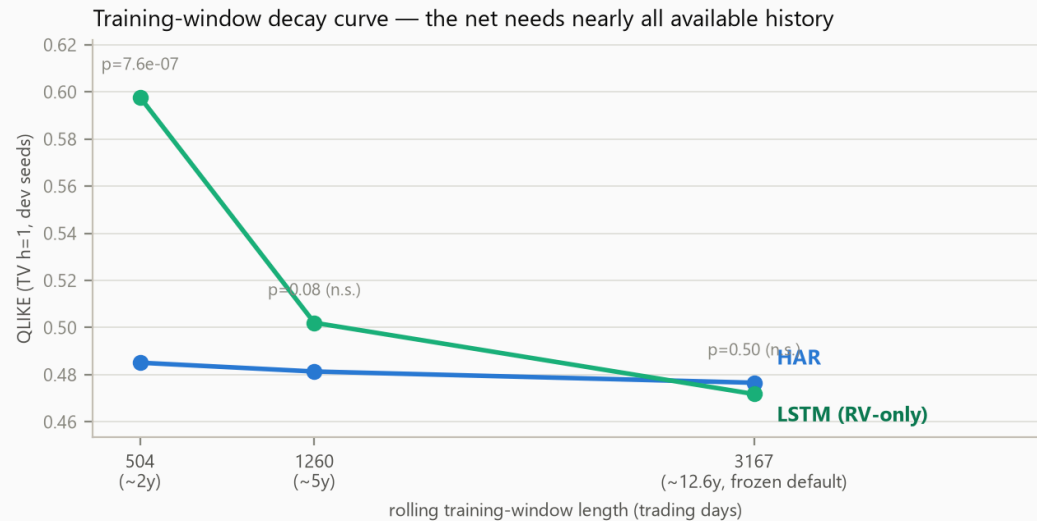


FIG. 3 — QLIKE VS. TRAINING-WINDOW LENGTH (RV-ONLY, TV H=1)

- At **~2 years** of history, HAR wins decisively ($p = 7.6 \times 10^{-7}$).
- At **~5 years**, HAR still leads ($p = 0.084$).
- Only at the **full ~12.6-year** frozen window does the network reach parity with HAR.
- Practical reading: don't expect this edge on short histories.

What this result doesn't claim

- **Multiplicity** — many DM tests are reported; all of them together, and the headline p-values sit far below any reasonable correction.
- **Statistical, not economic** — no transaction-cost or vol-targeting backtest; one asset (SPY) only.
- **Proxy-scoped, floor not ceiling** — TV claims speak to total daily variance, OC to the trading session only; this is a lower bound on extractable signal, not a ceiling.

THE FLOOR CLAIM

There is economically meaningful predictive information beyond HAR — at least 17% of QLIKE — and it lives in the auxiliary market-state features, not the deep architecture.

RV-onlya genuine null ($p = 0.585 / 0.667$)**RV+aux**decisive, every target, horizon, regime ($p \leq 1.9 \times 10^{-5}$)

Forecasts against realized volatility, full test span

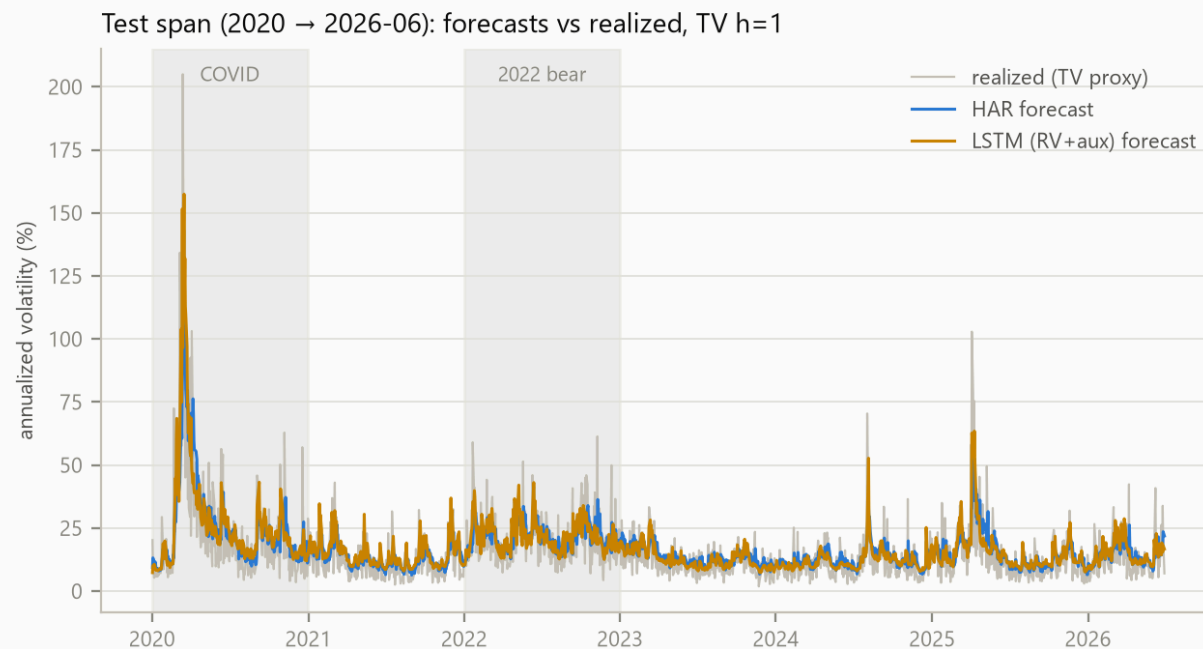


FIG. 4 — RV+AUX LSTM AND HAR FORECASTS VS. REALIZED VOLATILITY, 2020-2026